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Unit – 1 : Session – 12 : SLO – 2 | Logic gates (combined)

■ II. Multiple Choice Questions

1. b — AND
2. b — $Y = A \oplus B$
3. c — NAND
4. b — Inputs are the same
5. b — Digital comparator

■ I. Match the Following

1. AND gate logic symbol — b) $Y = A \oplus B$
2. OR gate output when both inputs are 0 — 0 (LOW)
3. NOT gate Boolean expression — c) $Y = A \oplus B$
4. NAND gate function — d) Universal gate for implementing AND, OR, NOT
5. EX-NOR gate application — e) Digital comparators and equality checking

■ III. Activity Questions

1. AND/OR/NOT circuit example: $F = (A \cdot B) + C$ — AND of A,B ORed with NOT of C. Truth table: $F=1$ whenever $A=B=1$ or $C=0$. Symbols drawn as D-shape (AND), curved (OR), triangle+bubble (NOT).
2. NAND: $Y = (A \cdot B)'$ — output 0 only when both inputs are 1. Universal implementations: NOT = A NAND A; AND = (A NAND B) NAND (A NAND B); OR = (A NAND A) NAND (B NAND B).
3. EX-NOR: $Y = (A \oplus B)'$ — output 1 when inputs are equal. Truth table: $00 \rightarrow 1, 01 \rightarrow 0, 10 \rightarrow 0, 11 \rightarrow 1$. Applications: digital comparators, equality checking, parity checking.